United International University

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Project Title: AI-Powered HEXAPOD for Autonomous and Wireless-Controlled Navigation

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Course Name: Electronics Laboratory

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Section: C

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Abstract:

Hexapod robots have gained significant attention in robotics due to their stability, maneuverability, and adaptability to rough terrains. This project presents the design and implementation of an AI-powered Spider Robot capable of autonomous movement, obstacle avoidance, and wireless manual control. The robot is controlled by an **ESP32** microcontroller and utilizes **MG90S** servo motors for leg movement. It follows a ripple gait pattern for smooth and efficient locomotion. To enhance navigation, the robot employs an **HC-SR04** ultrasonic sensor for obstacle detection and a **BMI160 IMU** sensor for self-balancing. Additionally, wireless communication is facilitated through an **NRF24L01** or **HC-12 RF** module, allowing manual control when needed. The system is powered by a 7.4V LiPo battery, ensuring sufficient energy for extended operation. The project aims to develop an intelligent, cost-effective, and scalable robotic platform suitable for surveillance, exploration, and research applications.

Project Overview:

The AI-powered Spider Robot is a six-legged hexapod designed for autonomous and wireless-controlled operations. With advancements in robotics and artificial intelligence, legged robots have proven superior to wheeled robots in navigating uneven terrains. This project focuses on building a cost-effective, efficient, and intelligent robotic platform that can move autonomously, avoid obstacles, and maintain balance while also allowing manual control through wireless communication which is an in-built mobile remote control.

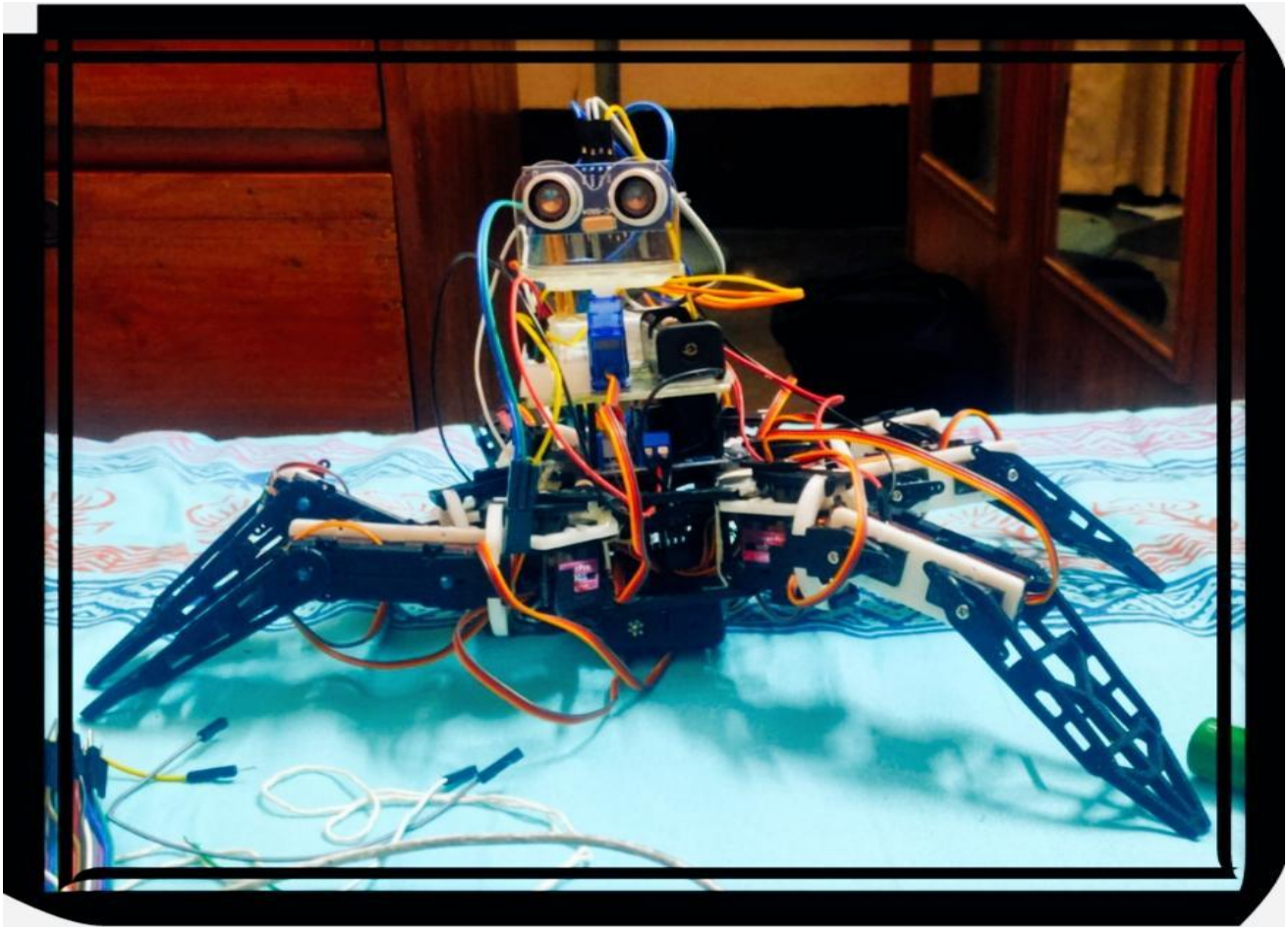


Fig-1: Hexapod

Objectives:

- Develop a hexapod robot with smooth and stable movement.
- Implement obstacle detection and avoidance using an ultrasonic sensor.
- Integrate self-balancing capabilities for improved stability.
- Enable wireless control via an RF module.
- Optimize power efficiency using a 7.4V LiPo battery.
- Enhance usability for applications such as surveillance, exploration, and research.

Features of the Hexapod Robot:

1. AI-Powered Autonomous Movement
Artificial Intelligent robots can efficiently navigate uneven terrain without the assistance of a human and had proven superior over the wheeled robots.

2. Wireless Control for Manual Operation
This Spider can be controlled through Mobile Application via Bluetooth.
HC-12 RF module enables long-range wireless control Ability to switch between autonomous and manual modes seamlessly.

3. Self-Balancing Mechanism
BMI 160 IMU sensor detects tilt and movement to maintain stability.
The ESP32 processes balance data and adjusts MG90S servo motors for correction.

4. Advanced Hardware Integration

ESP32 microcontroller with Wi-Fi & Bluetooth for processing and control.
MG90S servo motors provide precise movement for six legs.

5. 7.4V LiPo battery

7.4 V lipo battery for reliable and extended power supply.

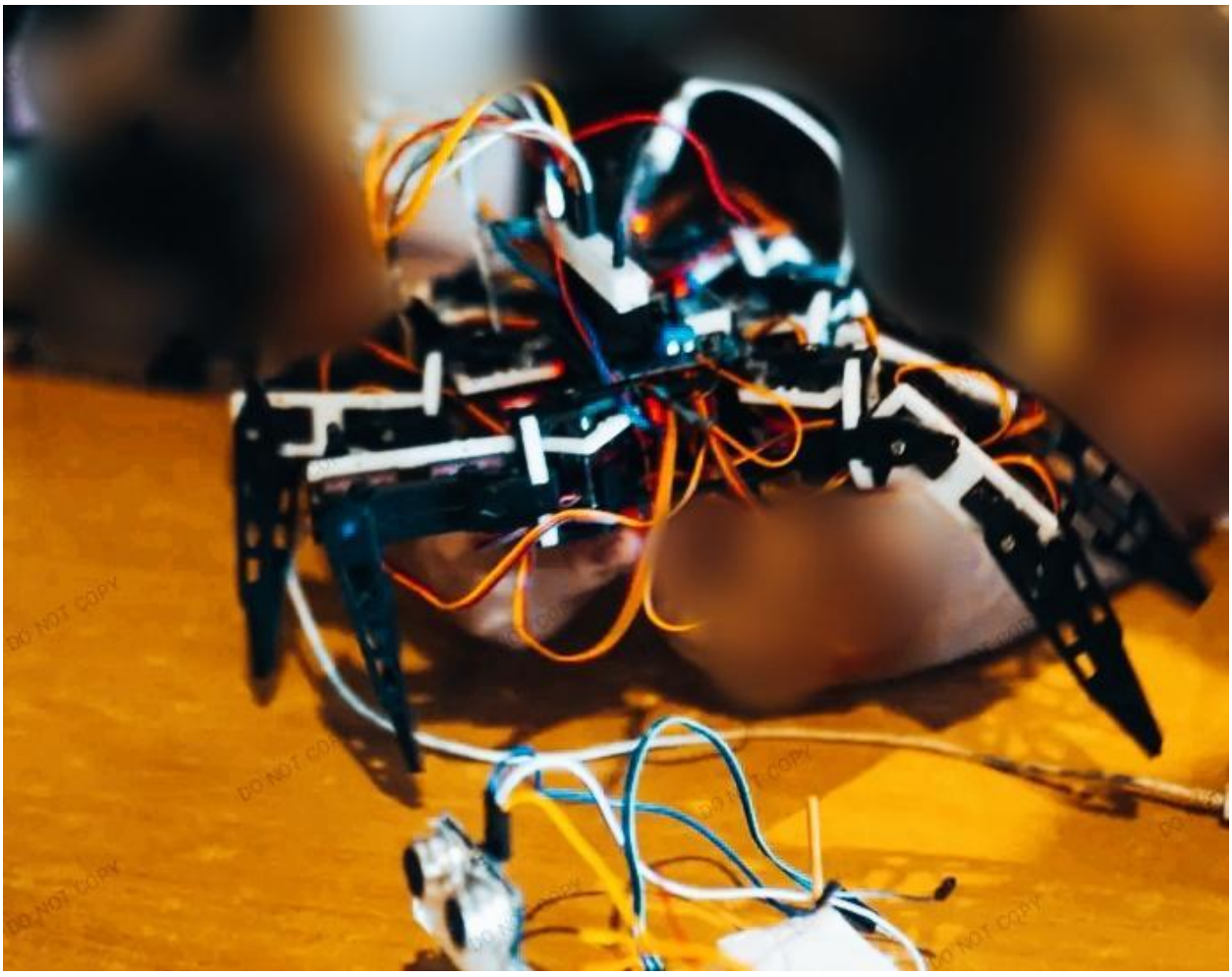


Fig-2: Hexapod

6. High-Performance Locomotion
Uses independent leg control for smooth, insect-like movement.
Optimized for speed, agility, and flexibility in various terrains.

7. Eye-Catching Design
Attractive and unique robotic structure for grabbing attention.
Innovative AI, self-balancing, and wireless features give a competitive edge.

8. Usability
Surveillance, exploration, and research are the main priorities of the Spider

Hardware Components:

1. **ESP32** – Main controller for processing and communication.
2. **MG90S Servo Motors** – For precise leg movement.
3. **HC-SR04 Ultrasonic Sensor** – For obstacle detection.
4. **BMI160 IMU Sensor** – For self-balancing.
5. **NRF24L01 or HC-12 RF Module** – For wireless manual control.
6. **7.4V LiPo Battery –** Power supply for the system.
7. **TXB0108/BSS138 Logic Level Shifter** – For voltage compatibility.

ESP32 OVERVIEW:

The ESP32 is a powerful and versatile microcontroller developed by Espressif Systems, widely used in IoT, robotics, and embedded systems due to its high performance, low power consumption, and built-in wireless connectivity. It features a dual-core or single-core Tensilica Xtensa LX6 processor with a clock speed of up to 240 MHz, making it significantly more powerful than traditional microcontrollers like the Arduino. With 520 KB of SRAM and support for external flash storage up to 16 MB, the ESP32 is well-suited for complex applications requiring data storage and processing.

One of its key advantages is integrated Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 + BLE, enabling seamless communication with other devices without additional modules.Due to its high processing power, built-in wireless features, and affordability, the ESP32 is an excellent choice for projects requiring efficient motor control, real-time sensor data processing, and wireless communication.

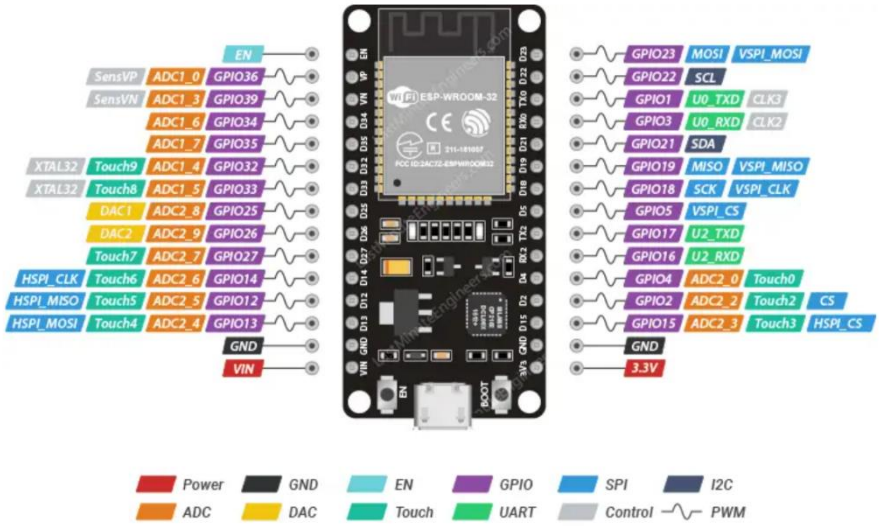


Fig-3: ESP-32

SERVO MG90S OVERVIEW:

The MG90S is a small, lightweight, and reliable digital servo motor widely used in robotics, RC models, and mechanical applications. It provides a moderate torque of 2.2 kg·cm at 4.8V and 2.5 kg·cm at 6.0V, which makes it suitable for lightweight robots and small mechanical systems. The servo motor is capable of fast movement with a speed of 0.1 sec/60° at 4.8V, ensuring quick and precise adjustments. It operates within a voltage range of 4.8V to 6.0V, making it compatible with typical power supplies used in small robots. The MG90S weighs only 13.5g and has compact dimensions (22.8mm x 12.2mm x 29.5mm), which is advantageous for space-constrained six-legged robots like hexapods.

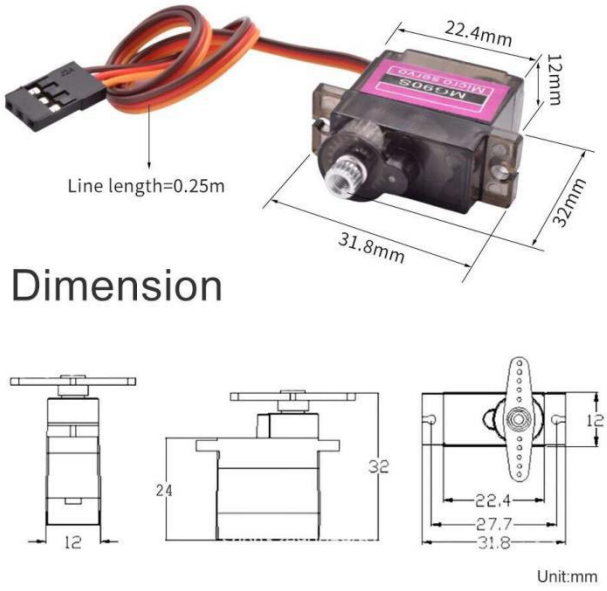


Fig-4: Servo-MG90S

HC-SR04 (ULTRA-SONIC SENSOR) OVERVIEW:

The HC-SR04 is a widely used ultrasonic distance sensor designed to measure the distance to an object using sound waves. It emits ultrasonic pulses at 40 kHz and measures the time it takes for the echo to return after bouncing off an object. This time is then used to calculate distance accurately.

It features four pins: VCC, GND, Trigger, and Echo. The sensor operates on 5V, and provides non-contact range detection from 2 cm to 400 cm with an accuracy of about ±3 mm.

Due to its low cost, simplicity, and reliable performance, the HC-SR04 is popular in applications like robotics, obstacle avoidance, distance measurement, and automation systems.



Fig-5: HC-SR04

BMI160 IMU SENSOR OVERVIEW:

The BMI160 is a compact and low-power 9-axis IMU (Inertial Measurement Unit) sensor developed by Bosch Sensortec, which integrates a 3-axis accelerometer, a 3-axis gyroscope, and a 3-axis magnetometer. This combination allows the sensor to provide accurate motion and orientation data, making it ideal for applications in robotics, wearables, and drones. The accelerometer measures acceleration along the X, Y, and Z axes, with selectable ranges of ±2g to ±16g, while the gyroscope

provides 3-axis rotational data with selectable ranges from ±125°/s to ±2000°/s. The integrated magnetometer offers 3-axis magnetic field data, helping with heading and orientation determination. The BMI160 is designed for low power consumption, drawing about 1.6 mA of current on average, making it suitable for battery-powered devices. It supports I2C or SPI communication protocols, enabling easy integration with various microcontrollers like Arduino and ESP32. With flexible output data rates ranging from 0.78 Hz to 1600 Hz, it can be tailored for both real-time data processing and low-power applications. The BMI160 is commonly used in robotics for motion tracking and stability, in wearable devices for detecting motion and orientation, and in drones for flight control and stabilization.



Fig-6: BMI160

HC-12 OVERVIEW:

The HC-12 is a long-range, low-power 433 MHz wireless RF module designed for reliable serial communication over extended distances. It operates on the ISM (Industrial, Scientific, and Medical) band at 433 MHz, making it suitable for applications requiring stable and interference-free communication. With a maximum range of 1.8 km in open areas (using the default 20 dBm transmission power and a proper antenna), the HC-12 is ideal for long-distance wireless control, sensor networks, and remote data transmission.

This module supports UART (serial) communication, making it easy to interface with microcontrollers like ESP32, Arduino, and Raspberry Pi without the need for complex communication protocols. It offers adjustable baud rates from 1,200 bps to

115,200 bps, allowing flexibility in data transmission speed based on the application. Additionally, the HC-12 supports Fu (Function) modes, which enable users to optimize the module for either longer range or higher data speed.

Operating on a 3.2V to 5V power supply, the module is efficient in power consumption, making it suitable for battery-powered projects. The low-frequency 433 MHz signal provides better obstacle penetration compared to 2.4 GHz RF modules like the NRF24L01, ensuring a more stable connection in environments with walls or interference.

The HC-12 is widely used in remote control systems, wireless sensors, robotic communication, and telemetry applications. In the Spider Robot project, it serves as an effective choice for wireless manual control, ensuring long-range and stable connectivity, even in outdoor or obstacle-rich environments.



Fig-7: HC-12

Implementation:

Powered Hexapod Spider Robot was developed using an integrated hardware-software approach to achieve intelligent autonomous movement, wireless control, and real-time environmental interaction. The implementation can be categorized into several key subsystems:

1. Control System

At the core of the robot is the ESP32 microcontroller, chosen for its dual-core processing capabilities, built-in Wi-Fi/Bluetooth, and compatibility with TinyML models. It coordinates all sensor inputs, motor outputs, and communication protocols.

1.Programming Environment: Arduino IDE

2.Languages Used: C/C++

3.Key Libraries:

→ ServoESP32.h – for controlling multiple servos

→ Wire.h and Adafruit_BMI160.h – for IMU data

→ RF24.h or SoftwareSerial – for wireless communication

→ TensorFlow Lite for Microcontrollers – for AI/ML logic (if ML applied)

2. Locomotion and Servo Control

The robot uses 12 MG90S servo motors, two per leg (coxa and femur), to enable spider-like movement through synchronized gaits.

Gait Mechanism:
Tripod gait implemented for smooth and stable walking.

Servo Calibration:
Each servo is calibrated to a neutral 90° position for balance and consistent movement.

Control Logic:
Custom PWM sequences determine walking, turning, or halting.

3. Obstacle Detection and Avoidance

An HC-SR04 ultrasonic sensor is mounted on the front of the chassis to detect nearby obstacles.

Detection Range: 2 cm to 400 cm

Response: If an obstacle is detected within a predefined range (e.g., < 20 cm), the robot either stops or re-routes using programmed evasive maneuvers**.**

Optional AI Enhancement: In autonomous mode, obstacle data can be fed into a TinyML decision-making model to select optimal direction based on past patterns.

4. Wireless Communication

The robot supports manual control via long-range wireless modules:

Modules Used:

NRF24L01 (2.4GHz wireless) or

HC-12 (433MHz serial RF)

Controller:

A separate ESP32/Arduino-based handheld remote sends directional commands using joystick or button inputs.

Modes:

1. **Manual Mode:** Direct control via remote
2. **Autonomous Mode:** Robot navigates based on sensor input and AI logic

MIT App Inventor:

To enable wireless manual control of the hexapod robot, we developed a custom mobile application using MIT App Inventor.

Since the robot uses an ESP32 microcontroller for onboard processing and Bluetooth communication, the app was designed to send specific control commands over Bluetooth. These commands were received by the ESP32, which then interfaced with the PCA9685 servo driver to control multiple MG90S servo motors. This setup allowed for precise and smooth leg movements, enabling real-time control of the robot's gait and direction via a smartphone.

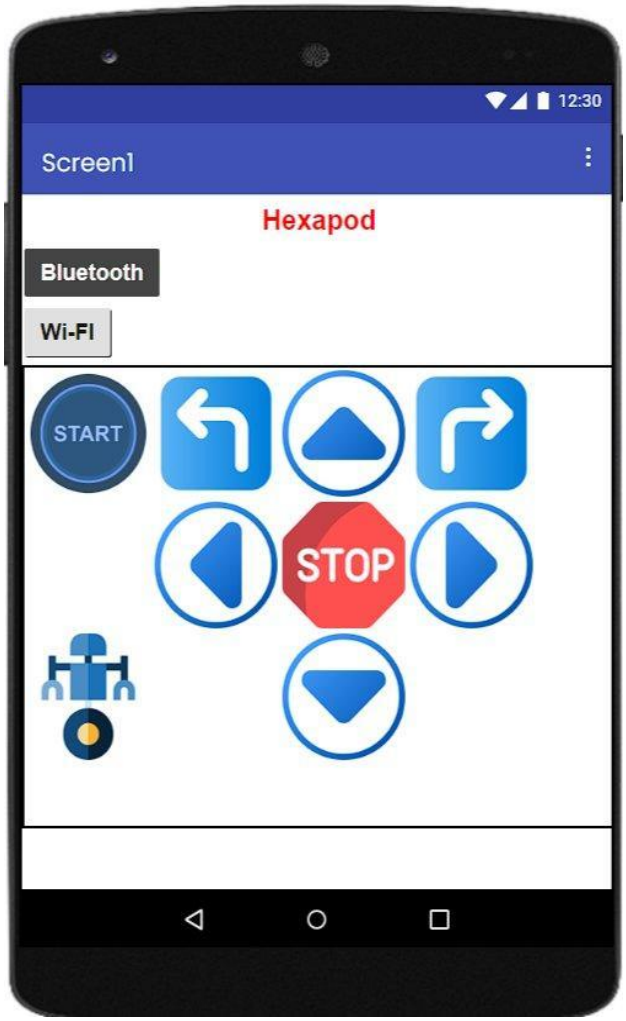


Fig-8: MIT Mobile Controller

5. Self-Balancing Mechanism

To enhance stability on uneven surfaces, a BMI160 6-axis IMU (gyroscope + accelerometer) is used:

Functionality:

1.Monitors pitch and roll angles

2.Detects tilts and adjusts leg positions dynamically

Algorithm:

A PID-like feedback system processes the IMU data to shift the servos slightly in real-time, improving balance during motion or obstacle climbing.

6. Power Management

The robot is powered by a 7.4V LiPo battery to support the high current demands of the servos.

1. **Voltage Regulation:** Buck Converter steps down to 5V for servos.3.3V LDO Regulator for ESP32 and sensors

2. **Power Isolation:** Servos and control electronics are powered from separate rails to prevent brownout resets during movement.

This modular and scalable design ensures that the robot is not only functional but also adaptable for future enhancements, such as:

→Voice control

→Computer vision with ESP32-CAM

→Cloud-based telemetry

Problems We Faced and How We Solved It

Usage of 7.4V Battery:

As we used 7.4 V LiPo Battery to power our system, it created a power overshot. Components like the ESP32 (3.3V), PCA9685 (5V) and MG90S servos (4.8-6V) require lower voltages. To prevent damage, we used Buck Converter, which regulated the voltage to provide constant 5V power.



Fig-9: LiPo 7.4V Battery

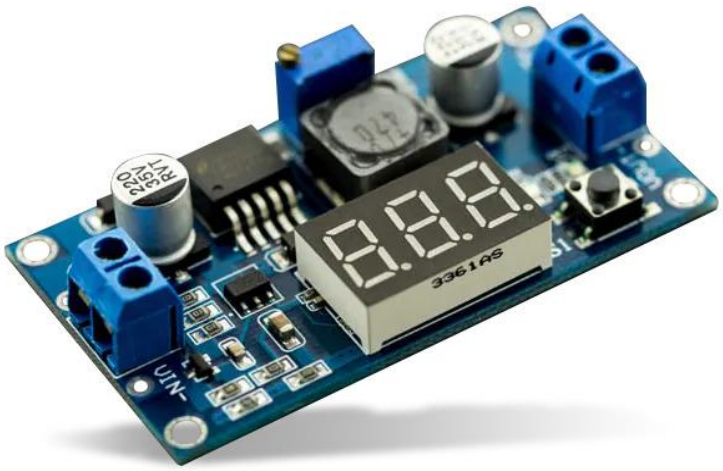
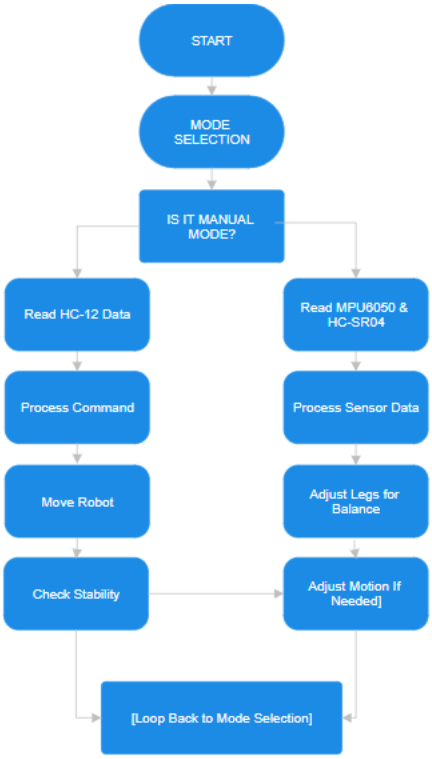


Fig-10: Buck Converter

Flowchart:



Code Segment:

```
#include <BluetoothSerial.h>
#include <Servo.h>

BluetoothSerial SerialBT;

// 12 Servos (6 legs × 2)
Servo leg1_hip, leg1_knee;
Servo leg2_hip, leg2_knee;
Servo leg3_hip, leg3_knee;
Servo leg4_hip, leg4_knee;
Servo leg5_hip, leg5_knee;
Servo leg6_hip, leg6_knee;

// Ultrasonic
#define trigPin 5
```

```
#define echoPin 4
long duration;
int distance;

int getDistance() {
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
duration = pulseIn(echoPin, HIGH);
distance = duration * 0.034 / 2;
return distance;
}

void stopMovement() {
leg1_hip.write(90); leg1_knee.write(90);
leg2_hip.write(90); leg2_knee.write(90);
leg3_hip.write(90); leg3_knee.write(90);
leg4_hip.write(90); leg4_knee.write(90);
leg5_hip.write(90); leg5_knee.write(90);
leg6_hip.write(90); leg6_knee.write(90);
}

void moveForward() {
leg1_knee.write(60); leg4_knee.write(60); leg5_knee.write(60);
delay(150);
leg1_hip.write(70); leg4_hip.write(70); leg5_hip.write(70);
delay(150);
leg1_knee.write(90); leg4_knee.write(90); leg5_knee.write(90);
delay(150);

leg2_knee.write(60); leg3_knee.write(60); leg6_knee.write(60);
delay(150);
leg2_hip.write(110); leg3_hip.write(110); leg6_hip.write(110);
delay(150);
leg2_knee.write(90); leg3_knee.write(90); leg6_knee.write(90);
delay(150);
```

```
leg1_hip.write(90); leg4_hip.write(90); leg5_hip.write(90);
leg2_hip.write(90); leg3_hip.write(90); leg6_hip.write(90);
}

void moveBackward() {
leg1_knee.write(60); leg4_knee.write(60); leg5_knee.write(60);
delay(150);
leg1_hip.write(110); leg4_hip.write(110); leg5_hip.write(110);
delay(150);
leg1_knee.write(90); leg4_knee.write(90); leg5_knee.write(90);
delay(150);

leg2_knee.write(60); leg3_knee.write(60); leg6_knee.write(60);
delay(150);
leg2_hip.write(70); leg3_hip.write(70); leg6_hip.write(70);
delay(150);
leg2_knee.write(90); leg3_knee.write(90); leg6_knee.write(90);
delay(150);

leg1_hip.write(90); leg2_hip.write(90); leg3_hip.write(90);
leg4_hip.write(90); leg5_hip.write(90); leg6_hip.write(90);
}

void turnLeft() {
leg1_hip.write(110); leg2_hip.write(110); leg3_hip.write(110);
leg4_hip.write(70);  leg5_hip.write(70);  leg6_hip.write(70);
delay(300);
stopMovement();
}

void turnRight() {
leg1_hip.write(70);  leg2_hip.write(70);  leg3_hip.write(70);
leg4_hip.write(110); leg5_hip.write(110); leg6_hip.write(110);
delay(300);
stopMovement();
}

void setup() {
Serial.begin(115200);
```

```
SerialBT.begin("HexapodESP32");

leg1_hip.attach(13); leg1_knee.attach(12);
leg2_hip.attach(14); leg2_knee.attach(27);
leg3_hip.attach(26); leg3_knee.attach(25);
leg4_hip.attach(33); leg4_knee.attach(32);
leg5_hip.attach(22); leg5_knee.attach(21);
leg6_hip.attach(19); leg6_knee.attach(18);

pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);

stopMovement();
}

void loop() {
int d = getDistance();
if (d < 15) {
stopMovement();
return;
}

if (SerialBT.available()) {
char c = SerialBT.read();
if (c == 'F') moveForward();
else if (c == 'B') moveBackward();
else if (c == 'L') turnLeft();
else if (c == 'R') turnRight();
else if (c == 'S') stopMovement();
}
}
```

LIMITATIONS

There are some limitations in our hexapod. Though it is designed as it can easily go to any place to do search operations. It can go most of the places but it fails to go in slimy areas. Because of the weight it drowns. It also can't operate in desert or sandy areas due to its pointed out leg, it will get stuck in sand. Its electric components are open so there is a chance of short circuit when it gets touched with water or the same kind of elements. It faces problems when it walks on slippery surfaces but it can be fixed by adding rubber pads in the legs. Everything has its own limitations. Our project is not out of it. We made it as a prototype, the limitation can be overcome though further updates.

FUTURE IMPROVEMENTS

This robot has been designed to solve some critical problems which are manually very hard and risky for human beings. We can upgrade the Spider Robot by enhancing its features like -

Climbing Ability: This robot can climb any obstacle or object and balance on rough surfaces as well. By adding suction cups, we can enhance its ability so that it can climb any building.

Camera: We can add a camera on it so that we can monitor the affected places. SOC [System on Chip] could be used to control the camera for monitoring the affected area and give the information by video capturing footage. For instance, when the hexapod robot will climb and walk through the affected areas, the camera will be capturing the affected moment and broadcast by live streaming on YouTube with the help of the MQTT broker

Critical Angle Climbing: This improvement can be one of the most important improvements for this robot if we can walk the robot smoothly near about 60' angles of the surfaces with a good balance.

Adding Sensors: In risky areas, it can detect dark and light surfaces if we add infrared sensors, as well as if we add sound sensors, it can detect sound by using the

sound sensor module and reach there for human safety purposes where an earthquake, fire incidents will occur

Adding Mechanical Arm By putting a small mechanical arm with drill bit on the top surface of the robot can enhance its ability more. It could help in affected areas by engaging or disengaging any objects by using the little mechanical arm.

Conclusion :

Robotics has become a popular field of study in this modern era, Day-to-day lives of people are being attempted to be made as easy, safe and efficient as possible. That is why we wanted to create a small robot having the potential to minimize the risk factors for humans in certain situations and which will also be capable of many functions. As discussed in the 'Results and Discussion' section, the software used in this robot is used for a specific motor controller. Both of these products were developed by two Chinese companies. Its crawling ability gives an advantage where wheels are irrelevant to use. Which means obstacles that are difficult to overcome by a rover with wheels, can be crossed over using the Spider Bot. With limited finances, resources and time we tried our utmost and left almost no stones unturned looking to make the robot more significantly useful. Perhaps with better circumstances, we wish to add more features in the robot as we discussed in a section above.

The capabilities of the hexapod should not be only limited in devastated places. Its functions can be extended based on how the legs are used. If proper hardware supplies and funds are provided, this robot can be strengthened for more load carrying abilities and other attributes, Under sufficient facilities, the robot can even have wall climbing abilities activated to it for example. From military usage to many other work stations, hexapod has the potential to broaden its utilization. However, there is no end to perfection. Therefore, with thorough observation of results from experiments on the hexapod, scope for improvement can always be established.